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PAPER

Decoding inner speech with functional connectivity

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Abstract

*Background:*Inner-Speech (IS) based Brain-Computer Interfaces (BCIs) offer potential communication solutions for individuals with disabilities by decoding brain signals generated during speech imagination. While most IS-BCI systems rely on time-frequency EEG features, this study investigates functional connectivity—specifically, motif synchronization (MS)—to determine whether interactions between brain regions improve the discrimination of imagined words. *Methods*: We analyzed Electroencephalography (EEG) data from the 'Thinking Out Loud' dataset by Nieto *et al* 2022, involving 10 participants mentally simulating four Spanish words ('Up,' 'Down,' 'Left,' 'Right'). Connectivity matrices were derived using the MS method, and node metrics (strength, PageRank) were calculated across 11 frequency bands. A two-stage classification framework was implemented: a support vector machine (SVM) ranked features by discriminatory power, followed by 5-fold cross-validation to identify optimal feature combinations. *Results:* The proposed approach achieved an average classification accuracy of 43.7%, comparable to previously reported results on the same dataset. *Conclusions:* Motif synchronization–based functional connectivity and graph metrics provide informative features for imagined speech BCIs by capturing cross-regional brain interactions. Further work is needed to improve robustness and generalization across sessions and subjects.

1. Introduction

Brain-Computer Interfaces (BCIs) are systems that utilize brain signals to control external devices, enabling interaction with the environment without relying on conventional pathways of the central nervous system, such as muscular or hormonal pathways. Brain activity involves electrophysiological, neurochemical, and metabolic phenomena, quantifiable through various monitoring techniques. A BCI system measures a user's brain signals, extracts specific features, and translates them into commands for an application. These commands are then translated into actions, such as prosthetic movement, wheelchair navigation, robot control, cursor manipulation (McFarland and Wolpaw 2011), or direct electrical stimulation of muscles or the brain itself (Pfurtscheller *et al* 2006). BCIs can serve as assistive and rehabilitative technologies for individuals with severe motor disabilities resulting from conditions like amyotrophic lateral sclerosis (ALS), stroke, locked-in syndrome (LIS), or spinal cord injuries. Additionally, BCIs have been increasingly explored for applications such as emotional monitoring of patients (Widge *et al* 2014) and soldiers (Miranda *et al* 2015), monitoring driver wakefulness, and use in video games (Plass-Oude Bos *et al* 2010). One of the widely used techniques to measure brain signals in BCI systems is electroencephalography (EEG), chosen for its high temporal resolution, relatively low cost, and portability. EEG measures the electrical activity generated by the flow of charges in neural tissue (Lotte *et al* 2015). The amplitude of the EEG signal indicates the synchrony of activity among neurons beneath the electrodes, where high amplitude correlates with high synchrony and vice versa (Pfurtscheller and Lopes Da Silva 1999). A strategy under investigation for generating EEG signals in BCIs is Inner-Speech (IS), involving the mental rehearsal of a verbal task without its execution (Alderson-Day and Fernyhough 2015). IS-based studies explore the imagination of syllables (e.g., /ba/ or /ku/) and/or words (e.g., 'up' or 'down'). Classification experiments have demonstrated that signatures of imagined speech brainwaves can distinguish linguistic content with varying degrees of success. These signatures include differences in activities in the alpha (~8–10 Hz), beta (~12–30 Hz), and theta (~4–8 Hz) bands. In a two-syllable imagination experiment with four participants, signal classification accuracy ranged from 62%–87%, using the Hilbert Transform to extract wavelet envelopes in the mentioned frequency ranges, showing higher discriminative power in the beta range (Zmura *et al* 2009). A study (Brigham and Kumar 2010), also working with the imagination of two syllables, achieved accuracies ranging from 46% to 88% in a 7 subjects study where they used k-nearest neighbors (KNN) to classify autoregressive coefficients as features extracted from preprocessed EEG. Deng and colleagues explored the imagination of two syllables in three rhythms and got accuracies ranging from 19% to 22% for a 6 classes task, using Hilbert spectra and LDA (Deng *et al* 2010). Others (DaSalla *et al* 2009) analyzed the imagination of phonemes and tried to recognize three tasks: /a/, /u/, and rest, obtaining accuracies ranging from 68% to 79% using common spatial patterns (CSP) and a nonlinear support vector machine (SVM). Another group (Kim *et al* 2014), instead, worked with three vowels /a/, /i/, and /u/. Using multivariate empirical mode decomposition, CSP for feature extraction and linear discriminant analysis (LDA) for classification, accuracies around 70% were reported. Considering works where recognition of words has been investigated, experiments comparing electrode configurations (Wang *et al* 2013) showed similar accuracy (~70%) for classifying two Chinese characters using full-head (30 electrodes) versus languageregion-focused (15 electrodes) setups. Single-electrode designs (Salama *et al* 2014) achieved 57%–59% accuracy for 'yes/no' imagination across classifiers (SVM, LDA). Riemannian manifold metrics with MRVM (Nguyen *et al* 2018) yielded 50% (three short words) and 66% (two long words) accuracy. For larger tasks, the 'Imagined Speech' dataset (Pressel Coretto *et al* 2017) reported 19%–22% accuracy (vowels/words) with Random Forests, marginally improved to 25%–30% using CNNs (Cooney *et al* 2020).

The 'Thinking Out Loud' dataset (Nieto *et al* 2022), comprising EEG recording from 10 participants imagining four Spanish directional words, has been analyzed in multiple studies. A classification accuracy of 30% was achieved using a 2D CNN based on the EEGNet architecture (Lawhern *et al* 2016, Van Den Berg *et al* 2021). Bidirectional Long Short-Term Memory (BiLSTM) models applied to raw EEG data improved accuracy to 36% (Gasparini *et al* 2022). Small-World Networks (SWNet) combined with Spiking Neural Networks and CNN yielded 27.7% accuracy , while Inter-Trial Coherence (ITC) features enabled 56% and 60% accuracy with KNN and SVM classifiers, respectively (Lopez-Bernal *et al* 2024). Separately, replication of the paradigm using continuous wavelet transforms attained 88% classification accuracy (Dhananjaya *et al* 2024). Some researchers explored open-vobaculary scenarios where transformers were used to predict whole sentences. They reported classification accuracies of 40.1% (Duan *et al* 2023) and 45.2% (Liu *et al* 2024). In a noteworthy study (Abdulghani *et al* 2023), researchers achieved classification accuracies exceeding 95% for a four-word recognition task. Their approach leveraged wavelet scattering networks combined with LSTM classifiers and employed speech imagination intervals of up to 60 s, significantly longer than those used in previous studies, demonstrating enhanced performance in decoding imagined speech signals. In a following work, the same team of researchers enhanced their classification accuracies to 99.32% by using multi-wavelet features and SVM (Abdulghani *et al* 2024).

Recent work has explored alternative methodological directions for imagined speech decoding with encouraging results. Saha *et al* reported substantial performance improvements over traditional hand-crafted features using a hierarchical deep learning framework based on EEG covariance representations. (Wu *et al* 2024) demonstrated that adaptive LDA significantly improves decoding stability and real-time control performance by mitigating EEG non-stationarity. More recently, (Carvalho *et al* 2024) showed that delay differential analysis can yield competitive classification performance by capturing nonlinear dynamical structure in imagined speech EEG. Others (Mohamed Selim *et al* 2024) showed that gamified paradigms (GWAPs) can enhance participant engagement while preserving EEG signal quality.

In the context of classifying IS-related tasks, Functional Connectivity (FC)–based approaches have received comparatively limited attention. FC differs from traditional spectral analysis in that, rather than focusing solely on the decomposition of brain signals into frequency bands, it explicitly characterizes how different brain regions communicate and interact during cognitive activity. By mapping these interactions, FC can capture task-specific functional networks and circuits that are engaged during the imagination of speech.

A small number of studies have explored FC in imagined speech using phase- and amplitude-based connectivity measures. (Panachakel and Ramakrishnan 2021) employed phase synchronization metrics such as Mean Phase Coherence and Magnitude-Squared Coherence to construct connectivity matrices between EEG channels, which were subsequently used as inputs to deep learning models, achieving high

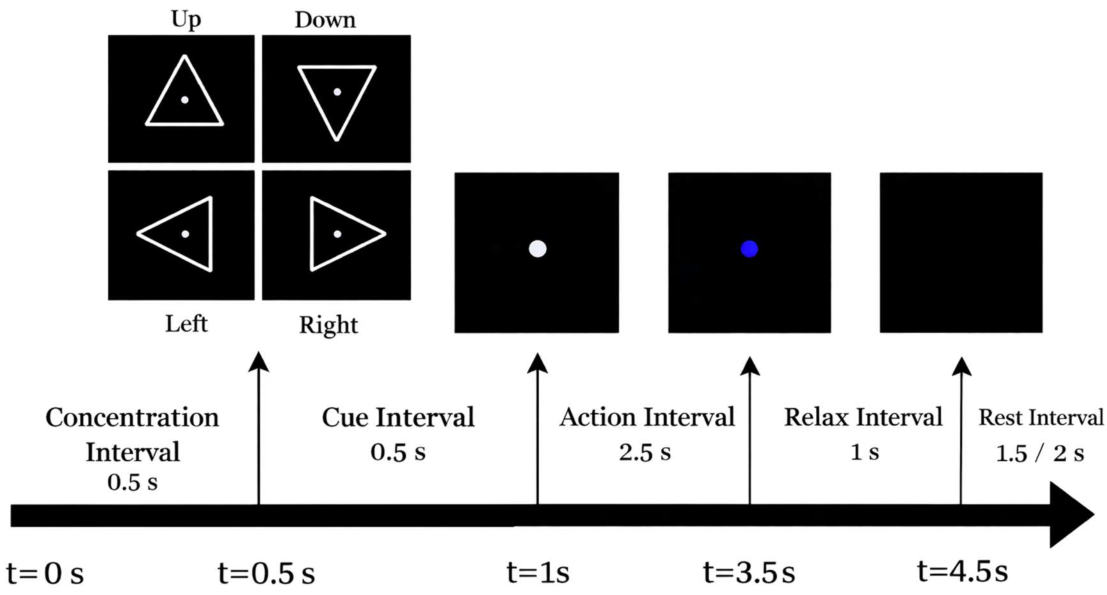

Figure 1. 'Thinking out loud' trial workflow.

classification performance in binary tasks. More recently, (Chengaiyan and Anandan 2022) combined phase synchronization measures, directed connectivity metrics, and entropy across multiple frequency bands as feature representations for deep learning–based classification.

In contrast to these frequency-domain and coupling-based approaches, the present work explores an FC framework based on the motif synchronization (MS) method (Rosário *et al* 2015). This method consists of comparing patterns of variations (motifs) in signals from different electrodes and has showcased favourable results in a distinct category of BCIs, namely those centered around motor imagery (Stefano Filho *et al* 2018, Vazquez *et al* 2023).

Given these considerations, this research aims to investigate the feasibility of using the MS method as a strategy for obtaining FC features and verify whether those facilitate the classification of EEG signals related to the imagination of speech of four words, using the 'Thinking Out Loud' dataset, mentioned earlier. We hypothesize that the use of information from the interaction of different brain regions might help discrimination of the imagined words.

2. Subjects, materials and methods

2.1. Thinking out loud dataset

2.1.1. Subjects

This study utilizes the 'Thinking Out Loud' dataset (Nieto *et al* 2022). The experimental procedures adhered to ethical guidelines and received approval from the 'Comité Asesor de Ética y Seguridad en el Trabajo Experimental' (CEySTE, CCT-CONICET, Santa Fe, Argentina, https://santafe.conicet.gov.ar/ceyste/). Ten right-handed participants, comprising four females and six males, with a mean age of 34 years (standard deviation = 10 years), and without hearing loss, speech impairment, or neurological, movement, or psychiatric disorders, willingly participated in the experiment after providing written informed consent. All participants were native Spanish speakers. None of the individuals had prior experience with BCI, and their involvement in the study encompassed approximately two hours of recording.

2.1.2. Experimental protocol

Each participant engaged in a single recording day, comprising three consecutive sessions. Between sessions, participants took self-selected breaks to alleviate potential boredom and fatigue. More details regarding the experimental setup can be found in (Nieto *et al* 2022).

The selected classes represented Spanish words ('arriba', 'abajo', 'derecho', 'izquierda') and were randomly presented in 200 trials for the first and second sessions. The trial count in the third session varied based on individual factors.

Figure 1 describes the composition of each trial, which commenced with a 0.5 s concentration interval, alerting participants about an upcoming visual cue. A white circle appeared, and participants were instructed to maintain focus until it disappeared at the trial's end. The cue interval introduced a white triangle pointing in a specific direction, initiating the 2.5 s action interval where participants imagined saying the corresponding word as if he/she was giving a direct order to the computer. The relaxation interval started after 2.5 s, marked by a blue circle, instructing participants to cease the activity without blinking until the circle disappeared at 4.5 s. Variable rest intervals, lasting 1.5 to 2 s, separated trials.

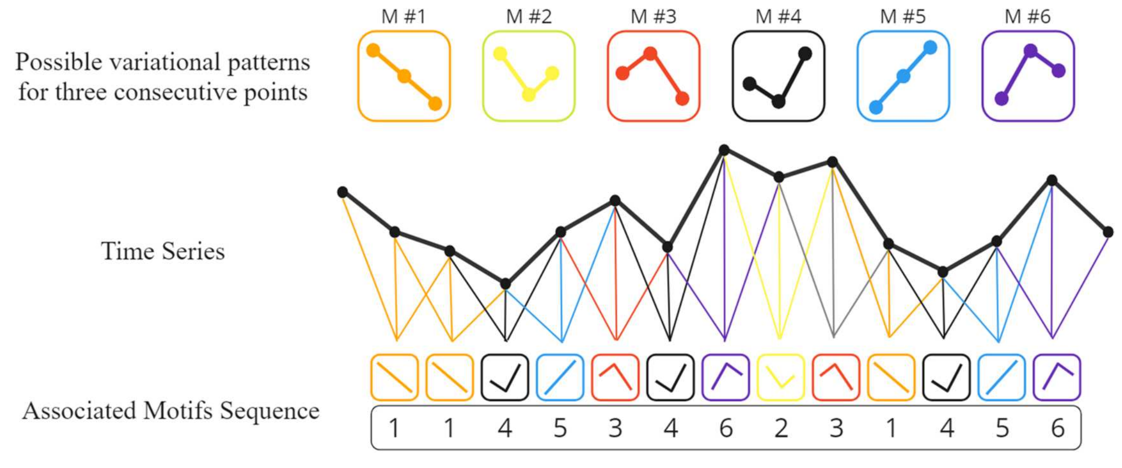

Figure 2. Three-point motifs method. M = Motif.

In addition, a concentration control was sporadically introduced during runs. Participants were asked, after specific trials, to identify the direction of the last presented class, providing an attention assessment. The protocol continued after participants responded by pressing any of the four arrow keys.

2.1.3. Thinking out loud data acquisition

In the process of data acquisition, the researchers used the BioSemi ActiveTwo high-resolution biopotential measuring system, with 128 active EEG channels, 8 external active EOG/EMG channels, and a sampling rate of 1024 Hz (Nieto *et al* 2022).

The EOG/EMG channels were labeled from EXG1 to EXG8, where EXG1 and EXG2 functioned as no-neural activity reference channels, placed in the left and right lobes of each ear; EXG3 and EXG4 were positioned over the participant's left and right temples to capture horizontal eye movements; EXG5 and EXG6 monitored vertical eye movements, particularly blinks, situated above and below the right eye, respectively; and finally, EXG7 and EXG8, placed over the superior and inferior right orbicularis oris, were used for tracking mouth movement during pronounced speech and ensuring no movement during inner speech and visualization condition runs (Nieto *et al* 2022).

2.1.4. Pre-processing

The data provided in the dataset was already preprocessed (Nieto *et al* 2022). Preprocessing steps consisted of:

1. Re-referencing: The MNE library's re-reference function was used, specifically utilizing channels EXG1 and EXG2. A virtual channel was created by averaging EXG1 and EXG2, and then this virtual channel was subtracted from each of the other acquired channels.
2. Digital Filtering: The data underwent comprehensive filtering using MNE functions, employing a zero-phase bandpass finite impulse response filter with lower and upper bounds set at 0.5 and 100 Hz, respectively, as well as a Notch filter at 50 Hz.
3. Epoching and Decimation: the sampling rate was reduced to 256 Hz and the continuous recorded data were epoched, preserving 4.5 s length signals that aligned with the temporal span between the initiation of the concentration interval and the conclusion of the relaxation interval. To structure the dataset, matrices of dimensions [channels $\times$ samples] for each trial were systematically stacked, resulting in a final tensor characterized by the dimensions [trials $\times$ channels $\times$ samples].
4. Independent Component Analysis (ICA): Infomax Independent Components Analysis (ICA) was applied to the EEG channels without the incorporation of Principal Component Analysis (PCA), resulting in the extraction of 128 sources. To refine the dataset, a correlation analysis with the EXG channels was employed, enabling the identification and subsequent exclusion of artifacts associated with blink, gaze, and mouth movement.

2.2. Functional connectivity: motif synchronization method

In this work we used the MS method to compute FC. This method, proposed in 2015 by Rosário *et al* (2015), has already been used in BCI related applications (Vazquez *et al* 2023). In simple terms, the motifs method is a qualitative approach that transforms EEG time series into new series, known as motifs series. These motifs series are generated by identifying how the original time series varies and then assigning labels to all possible behaviors that may occur. It is important to note that different numbers of points in an analysis will result in different possibilities of

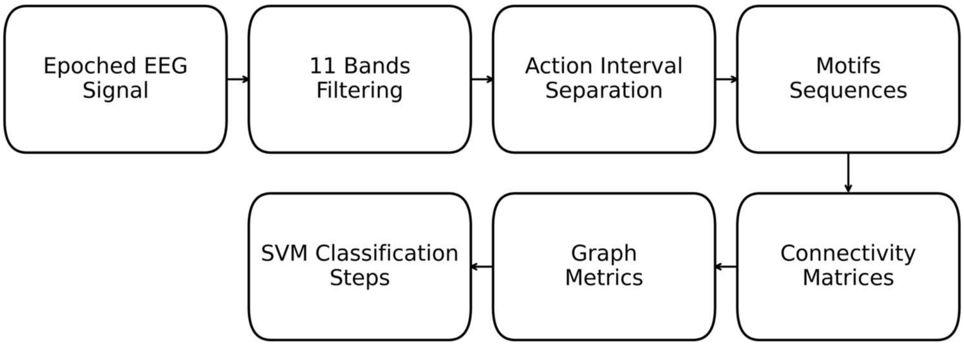

Figure 3. Proposed methodology.

patterns (motifs). This is illustrated in figure 2, which shows an example using 3-point motifs.

In figure 2, a window containing the desired number of points (three in this case) is moved along the EEG series, and at each step the window is assigned to one of the corresponding labels for possible patterns, thereby creating the motifs series from the EEG time series. It is important to note that, in this process, information about the exact variation between two points in time is not considered; only the general qualitative shape of the signal is considered. It should be noted that the length of the motifs series is always shorter than that of the original time series (specifically, if the number of points in the original series is N and the number of points in the motifs is n, the number of points in the motifs series is N-n+1).

Using the motifs approach, the similarity between two electrode time series is assessed based on the agreement between their motifs' series, determined by calculating the number of coincidences between the two sequences. Furthermore, it is possible to perform this process by introducing a specific shift in the second sequence, allowing the derivation of a 'lagged connectivity' measure. In this research, we utilized six different lag strategies to compare their discriminatory power, all of them utilized 3-point motifs, a sliding step of one sample and no sensitivity analysis was conducted:

1. Max (0, 1, 2, 3): This strategy involves computing the number of agreements between motifs sequences across shifts of 0, 1, 2, and 3 units. The resulting connectivity between electrodes is determined by selecting the maximum value among these agreements.
2. Max (1, 2, 3): Similar to the previous method, this approach considers shifts of 1, 2, and 3 units to calculate the number of agreements between motifs sequences, with the connectivity being established based on the highest value among these.
3. Lag 0: This strategy entails evaluating the number of agreements between motifs sequences with no temporal shift between them.
4. Lag 1: The analysis here involves calculating the number of agreements between motifs sequences with a 1-unit temporal shift.
5. Lag 2: Similar to Lag 1, this approach assesses the number of agreements between motifs sequences, but with a 2-unit temporal shift.
6. Lag 3: Building on the previous two strategies, Lag 3 involves determining the number of agreements between motifs sequences with a 3-unit temporal shift.

Except for the lag-0 case, all methods yield directed graphs. In this work, we preserve this directed structure.

The exploration of such methods is driven by the need to understand the impact of volume conduction (VC) on signal discrimination when working with the MS method. VC arises when the same brain activity is simultaneously recorded by multiple electrodes in different brain areas, leading to spurious correlations. As this effect is instantaneous, it is expected to influence zero lag connectivity measures.

The methodology devised for obtaining FC features and subsequently classifying samples based on the imagined word, represented in figure 3, can be outlined through the following steps:

1. Extraction of epochs associated with the imagination of each word using the library available at <bold>https://github.com/N-Nieto/Inner_Speech_Dataset</bold>.
2. Temporal series filtering across different frequency bands using the Scipy library in Python.
3. Computation of motif sequences.
4. Computation of connectivity matrices.
5. Calculation of graph metrics.

6. Two-stage classification using Support Vector Machines (SVM).

SVMs were chosen as classifiers due to their previous success in BCI applications (Costantini *et al* 2009, Stefano Filho *et al* 2018, Antony *et al* 2022, Vazquez *et al* 2023).

2.2.1. Extraction of epochs

To extract EEG data from participants, pre-partitioned into specific epochs associated with the activities under investigation, the functions 'Extract_data_multisubject' and 'Transform_for_classificator' from the Inner Speech Dataset library were utilized. These functions employed the MNE library in Python for obtaining the 2 s epochs corresponding to each section of speech imagination (note: the first 0.5 s of the 2.5 s of speech imagination was not used, to ensure task alignment). As a result of this process, data matrices in the format (n, 128, 512) were obtained: 512 time-samples for each one of the 128 electrodes for a total of n runs for each participant. Although the number of epochs varied across participants (180–240), all classes were balanced, and no additional balancing was required.

2.2.2. Temporal series filtering

Subsequently, the sosfilt and butter functions from the scipy.signal library were employed to calculate and apply filtering coefficients to the time series, allowing segmentation into 11 distinct frequency bands (12–15 Hz, 15–18 Hz, …, 42–45 Hz). This frequency range was selected to emphasize higher-frequency activity (high-beta/low-gamma), which is commonly associated with task-related neural processing in EEG-based decoding and BCI studies. Lower-frequency bands (theta and alpha) were excluded by design, as the objective was to focus on frequency ranges more directly linked to task execution rather than to provide a full-band spectral characterization of the signals. As a result of this process, data matrices in the format (n, 11, 128, 512) were obtained.

2.2.3. Computation of motifs sequences

Following the filtering of signals across different frequency bands, the sequences of motifs were computed for each filtered time series obtained in the preceding step. To accomplish this, a function was devised for motif calculation and subsequently applied along the temporal series.

2.2.4. Computation of connectivity matrices

From the motifs' sequences obtained in the previous step, connectivity matrices were constructed by systematically comparing motifs sequences between each pair of electrodes.

In this procedure, the connectivity metric is established based on the maximum number of matches identified between two motifs sequences (Rosário *et al* 2015), considering that the second sequence may be shifted by 0, 1, 2, or 3 points relative to the first. This metric quantifies the value of the connection between the electrodes associated with these sequences, enabling precise mapping and characterization of the functional relationships among different electrodes within the analysis context. At the end of this stage, matrices were normalized using min-max normalization.

2.2.5. Computation of graph measures

Using graph theory, connectivity matrices were analyzed to extract two key network measures for each node (electrode). The first, out-strength, reflects the total number of connections a node transmits, indicating its outgoing influence. The second, PageRank Centrality (PRC), quantifies a node's relative importance within the network. Developed by Larry Page in 1998 for Google's search engine (Brin and Page 1998), PRC simulates a random surfer navigating the web by following hyperlinks. In the context of our network, it estimates the likelihood of encountering a specific node during an extended random walk across the connections. The equation defining PRC is:

$$PR(i) = 1 - d + d\sum \frac{PR(j)}{L(j)}$$

In this equation, $\boldsymbol{PR(i)}$ represents the PRC of node i, while d is the damping factor, typically set between 0.85 and 0.95, which regulates the influence of previous PRC scores on the current iteration. A higher d value emphasizes the significance of incoming links. In this work, we used $d = 0.85$. The formula includes a summation term iterating over all nodes that have directed links pointing to the current node. This term computes the weighted contribution of PRC from each incoming neighbor. The weight is determined by dividing the PRC score $\boldsymbol{PR(j)}$ of the neighbor by the number of outgoing links $\boldsymbol{L(j)}$ from that neighbor. Essentially, a node with a high PRC and fewer outgoing links contributes more weight to the PRC of the current node.

Although originally designed for non-weighted graphs, PRC is inherently suited for weighted directed graphs, where the strength of connections can influence navigation. NetworkX, a Python package, was employed to compute both these metrics. Since PRC values are typically several orders of magnitude smaller than strength values, we applied a linear rescaling (by a factor of 100) to the PRC features to bring them to a comparable numerical range. This operation is equivalent to a change of units and does not affect the underlying feature relationships.

In summary, upon the completion of all processing steps, two graph measures were extracted for each node (corresponding to an electrode) in each of the eleven frequency bands that were analyzed.

Table 1. Accuracies obtained with each connectivity strategy.

Subject	Max (0,1,2,3) (%)	Max (1,2,3) (%)	Lag 0 (%)	Lag 1 (%)	Lag 2 (%)	Lag 3 (%)
S1	43.0	**48.0**	40.0	46.5	44.5	47.0
S2	42.9	**47.5**	40.0	46.7	45.0	45.4
S3	38.3	**45.6**	41.1	38.9	40.0	43.3
S4	35.8	**42.1**	39.6	37.9	32.1	35.4
S5	**42.5**	39.6	37.1	**42.5**	39.2	38.3
S6	42.6	**44.9**	44.4	43.1	45.8	33.8
S7	36.7	**42.4**	39.2	41.7	40.4	39.2
S8	48.0	**51.0**	46.5	50.0	49.5	49.5
S9	42.1	39.6	41.2	42.1	40.0	**46.2**
S10	37.5	36.3	37.9	35.4	36.2	**41.2**
Mean	40.9	**43.7**	40.7	42.5	41.4	42.3
Std	3.6	4.3	2.7	4.2	4.8	5.5

discriminability was first evaluated using stratified 5-fold cross-validation, where folds were constructed such that samples from all recording sessions were present in each fold (i.e., a mixed-session split). In this setting, training and testing sets were drawn from the same subject, and the cross-validation procedure assessed within-subject discriminability rather than cross-session or cross-subject generalization. Feature ranking and classification were performed separately for each subject, yielding an average classification accuracy per participant.

Subsequently, an incremental feature inclusion strategy was applied by progressively combining the top-ranked features in order to identify the feature subset that maximized the mean cross-validated accuracy for each subject. We note that feature ranking

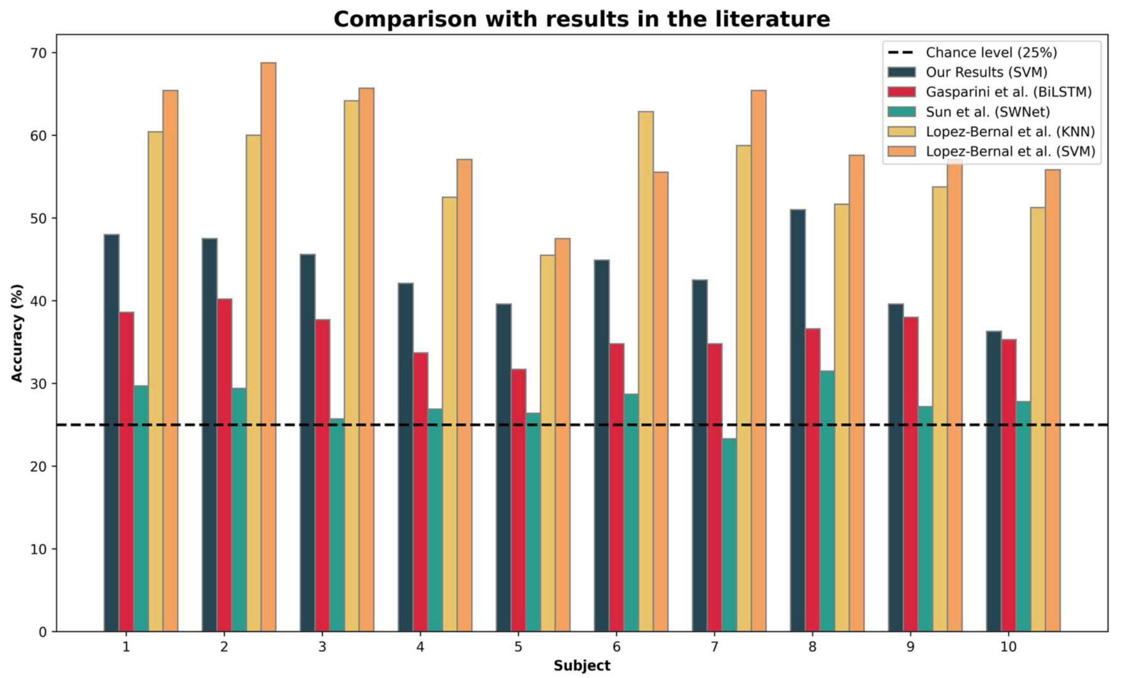

Figure 4. Comparison between the results obtained in the present study and those of (Gasparini *et al* 2022, Sun 2024) , and (Lopez-Bernal *et al* 2024), along with the chance level line of 25%.

2.3. Two stage classification process through SVM

The discriminative capacity of the features obtained in the processing steps was assessed using SVM, a classical method for data classification and which has already been successfully applied in the BCI context (Rong *et al* 2012, Vazquez *et al* 2023). The kernel used was the Radial Basis Function (RBF); default hyperparameters were adopted ($C = 1.0$, $\gamma =$ 'scale'), with no class weighting (class_weight = None), as classes were balanced for all participants after preprocessing. Model training followed the default LIBSVM stopping criteria (tolerance = 1e−3, maximum number of iterations = unlimited).

The classification procedure was performed independently for each subject using a within-subject validation design. For each participant, feature was performed prior to this incremental evaluation and was not recomputed within each cross-validation fold. Therefore, the feature-selection procedure is not fully nested and may introduce optimistic bias in performance estimates, which is acknowledged as a methodological limitation and motivates future work using leakage-free, nested validation frameworks.

3. Results

Table 1 displays the highest accuracies achieved for each of the connectivity strategies. The average accuracy of 45.8% surpasses those obtained by Gasparini *et al* and by Berg *et al*, who worked with the same dataset, which were lower than 37.0% (Van Den Berg *et al* 2021, Gasparini *et al* 2022), however, it falls short along with the chance level line of 25%.

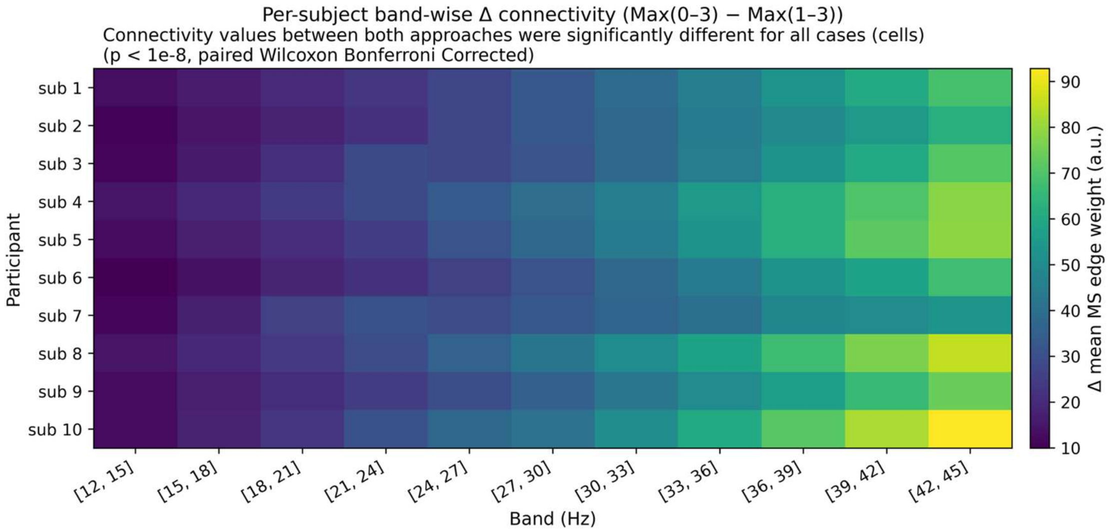

Figure 5. Per-subject band-wise effect of including zero-lag motifs on motif-synchronization connectivity. For each participant, frequency band and connectivity strategy (Max(0–3) and Max(1–3)), off-diagonal connectivity weights were summed for the connectivity matrix of each epoch, and then averaged over epochs. The heatmap shows, for each participant and frequency band, the difference in mean connectivity values between strategies ($\Delta = \mathrm{Max}(0\text{–}3) - \mathrm{Max}(1\text{–}3)$). Connectivity weights are expressed in arbitrary units, as they correspond to counts of motif-coincidence events. For all subject–band combinations, the increase in connectivity associated with including lag 0 was statistically significant ($p < 1\text{e-}8$, paired Wilcoxon, Bonferroni corrected).

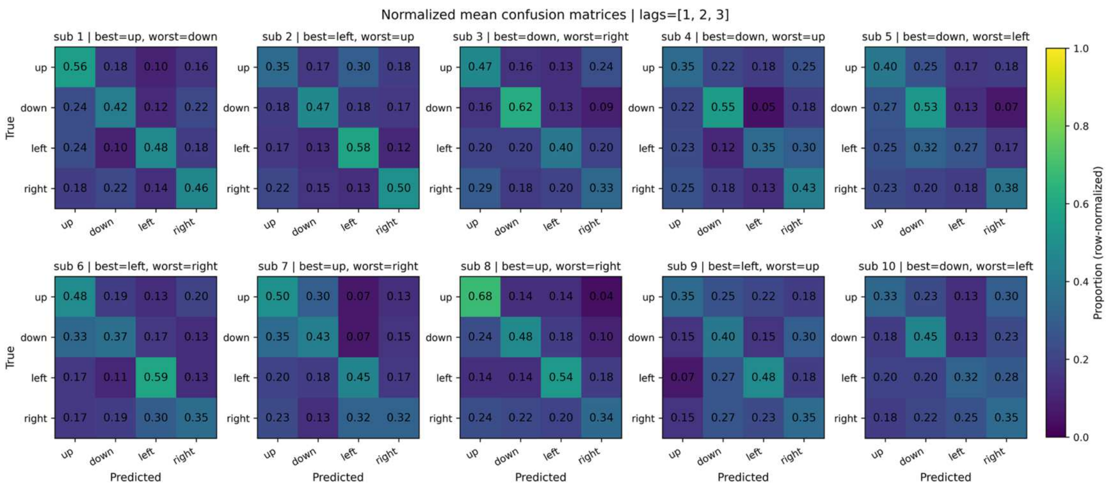

Figure 6. Subject-wise normalized confusion matrices for imagined speech classification using delayed motif synchronization (lags = [1, 2, 3]). Each subplot shows the row-normalized mean confusion matrix of an individual participant, computed across 5-fold stratified cross-validation. Titles indicate, for each subject, the class with the highest and lowest recall. Despite clear inter-subject variability, a consistent diagonal structure is observed across participants, indicating reliable class-wise discriminability when delayed functional interactions are considered.

of the results from (Lopez-Bernal *et al* 2024), who achieved average accuracies of 56.1% and 59.6% using KNN and SVM, respectively. In particular, figure 4 compares our results with those of (Gasparini *et al* 2022) and (Lopez-Bernal *et al* 2024), for each subject. Figure 5 examines the effect of including zero-lag interactions at the level of the connectivity matrices themselves, showing a systematic increase in motif-synchronization weights when lag 0 is included across subjects and frequency bands. This pattern is consistent with the expected inflation of instantaneous connectivity due to volume conduction and related mixing effects in EEG, supporting the choice of excluding zero-lag motifs in the subsequent classification analyses. Figures 6 and 7 provide a complementary view of performance through normalized confusion matrices, showing consistent diagonal structure across subjects and at the group level, despite variability in class-wise performance. Figures 8 and 9 further describe the distribution of selected features, indicating that electrode and frequency band contributions are spatially and spectrally

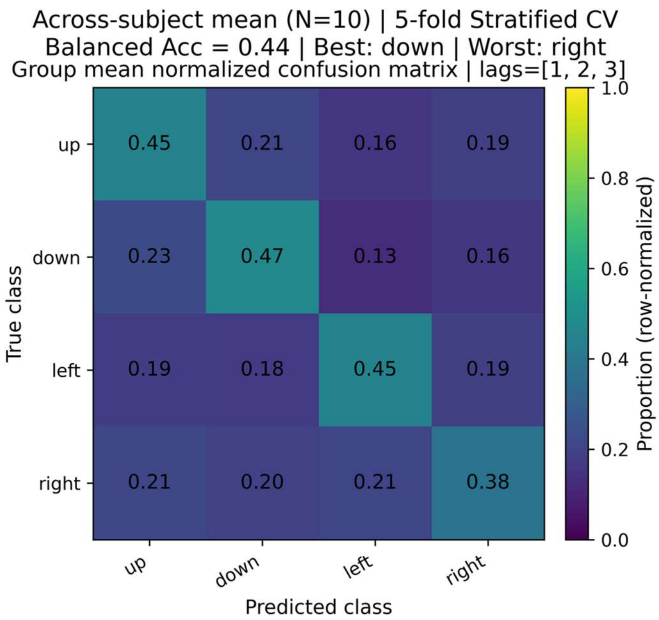

Figure 7. Group-level normalized confusion matrix for imagined speech classification using delayed motif synchronization (lags = [1, 2, 3]). The matrix represents the across-subject mean (N = 10) of row-normalized confusion matrices shown in figure 5. Diagonal elements correspond to class-wise recall, whose average defines the balanced accuracy (44%). The results show higher discriminability for the 'down' class and lower performance for the 'right' class. CV = cross-validation; Acc = accuracy.

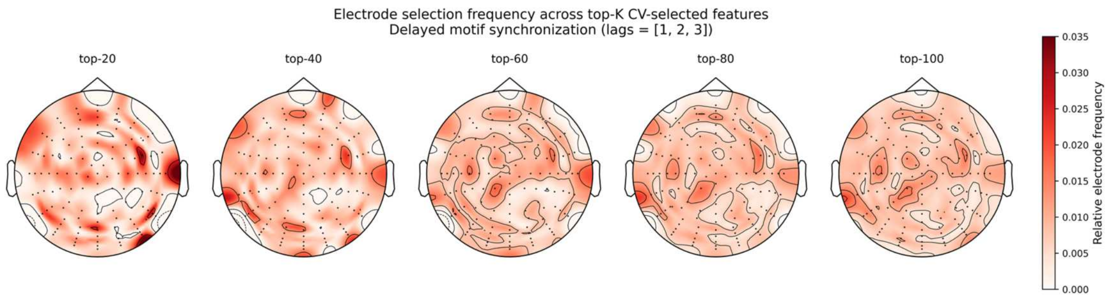

Figure 8. Topographic distribution of electrode selection frequency for increasing numbers of CV-selected features (top-20 to top-100) using delayed motif synchronization (lags = [1, 2, 3]). Each subplot shows the relative frequency with which electrodes appear among the top-K features aggregated across subjects and cross-validation folds. The spatial patterns remain largely stable as K increases, suggesting robustness of the identified functional connectivity contributions, with no strong localization to a single focal region.

distributed and remain relatively stable across different numbers of selected features.

4. Discussion

The aim of this work was to investigate whether functional connectivity features, obtained with the motif synchronization method (Rosário *et al* 2015), were able to improve the classification of EEG signals related to the imagination of speech of four words. To the best of our knowledge, this is the first time this type of features are used in the context of IS-BCIs. For this, two graph metrics (strength and PRC), obtained from six different functional connectivity strategies, were applied to the 'Thinking Out Loud' dataset (Nieto *et al* 2022). Within these strategies, four looked only at the relation within EEG time series considering a specific delay (lag) among them (from 0—no lag—to 3, i.e., 3 time points' lag, or around 12 ms delay), and two looked at the best match among a set of lags: 1 to 3, and 0 to 3. It is important to stress that a 0 (zero) delay match corresponds to signals measured simultaneously at two EEG locations, which has been associated with the volume conduction effect, i.e., a signal emitted by the same brain source that arrives simultaneously to both EEG sites due to conductivity of the surrounding tissues. This has been deemed as an artifact in EEG studies (e.g., (Peraza *et al* 2012, Brunner *et al* 2016, Ruiz-Gómez *et al* 2019)) given that, ideally, two different sites should measure signals arising from different brain sources, preferrably located right under the electrode sites.

The results obtained were similar among individual delays, with average accuracies ranging from (40.7 2.7)% for no lag to (42.5 4.2)% for the 1 time

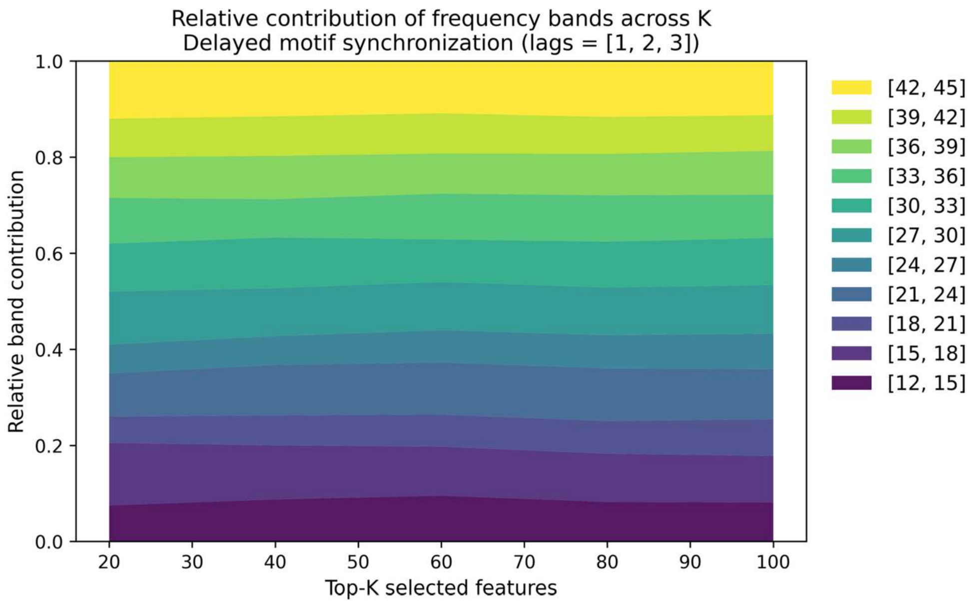

Figure 9. Relative contribution of frequency bands as a function of the number of selected features (K) for delayed motif synchronization (lags = [1, 2, 3]). The stacked area plot shows the proportional composition of frequency bands among the top-K features aggregated across subjects and cross-validation folds. The overall band distribution remains largely stable as K increases, indicating that the decoding performance relies on distributed spectral contributions rather than a single dominant band.

point (~4 ms) lag. However, average accuracies for the combined-lag strategies were slightly higher, reaching (43.7 4.3)% for the 1–3 lags and (40.9 3.8)% for the 0–3 lags. The results show that functional connectivity computed using delayed interactions (lags 1–3) consistently outperforms configurations that include zero-lag connectivity. This finding is consistent with the well-established view that zero-lag coupling in EEG is particularly susceptible to volume conduction, which can introduce spurious correlations that do not reflect genuine functional interactions. By excluding zero-lag motifs, the motif synchronization framework appears to emphasize delayed dependencies that are more likely to reflect true neural communication, leading to improved classification performance. These results support the interpretation that delayed functional interactions provide a more physiologically meaningful basis for imagined speech decoding. This behavior is further illustrated by the class-wise analysis shown in figures 5 and 6. Although overall accuracies remain moderate, the normalized confusion matrices reveal a consistent diagonal structure across subjects and at the group level, indicating that the performance gains obtained with delayed interactions are not driven by a single class but reflect systematic class-wise discriminability. At the same time, noticeable inter-subject variability highlights the inherent difficulty of the imagined speech task and the heterogeneity of neural representations across participants.

Compared to other studies in the literature, our method surpasses those obtained by two of the studies which used the same dataset (Van Den Berg *et al* 2021, Gasparini *et al* 2022), but achieves accuracies on average 11.8% lower than those reported by (Lopez-Bernal *et al* 2024), which are the highest accuracies reported so far with this dataset. Regarding the first two studies, both used deep learning models for classification, namely a 2D CNN based on the EEGNet (Lawhern *et al* 2016) in (Van Den Berg *et al* 2021) and a Bidirection Long Short-Term Memory (BiLSTM) in (Gasparini *et al* 2022), achieving 30% and 36% accuracy respectively, for the four-word classification task. Figure 4 shows a detailed comparison between our results, (Gasparini *et al* 2022, Sun 2024, Lopez-Bernal *et al* 2024) (individual results for the other study, (Van Den Berg *et al* 2021), were not provided). One reason why deep learning might not have succeeded is its dependency on extensive datasets to achieve optimal performance, which was not the case of the present dataset. Additionally, in the study by (Van Den Berg *et al* 2021), the data was left unprocessed (as in (Gasparini *et al* 2022)), and their model was trained exclusively using a subset of 26 electrodes situated on the left hemisphere, meaning they might have lost important information by disregarding the other 102 electrodes. It is plausible that the slight improvement in classification accuracy we obtained stems from the ability of FC to capture the interrelations between signals from various brain areas. This interpretation is supported by the feature distribution analyses shown in figures 7 and 8. The topographic maps suggest that discriminative information is not confined to a small subset of electrodes but is instead distributed across multiple channels, while the band-wise analysis indicates a relatively stable contribution of several frequency ranges across different numbers of selected features. Taken together, these observations are consistent with the notion that the proposed approach captures broadly distributed connectivity patterns, rather than relying on highly localized or narrowly band-specific effects. Regarding the third study, caution is needed when interpreting this comparison due to key methodological differences. In Lopez-Bernal *et al*'s work, which achieved accuracies of 56.1% and 59.6% using KNN and SVM respectively, classification was conducted on data from each recording session independently, effectively reducing the total number of samples per class by a factor of 3. Additionally, each sample in their study included 5,760 features, approximately twice the number of features used in our study. Although our classification accuracies were slightly lower, they were obtained using a more compact feature representation and a larger effective sample set per participant. This suggests a more parsimonious representation of the neural patterns associated with speech imagery, which may be advantageous for robustness and generalization, without explicitly evaluating cross-session performance.

In comparison with other studies in the field, our research focuses on the classification of EEG signals associated with the imagination of complete words. While our achieved classification accuracies hover around 45%, it is notable to highlight that although most of the referenced studies attained higher accuracies, differences in experimental paradigms must be considered. Most of these studies concentrated on simpler tasks, often involving the imagination of only two classes, which sometimes comprised syllables or short words. For instance, (DaSalla *et al* 2009) analyzed phoneme imagination tasks with accuracies ranging from 68% to 79%, while (Salama *et al* 2014) explored the discrimination of the imagination of two words, 'yes' and 'no', achieving accuracies between 57% to 59%. Moreover, several studies, such as (Kim *et al* 2014) and (Nguyen *et al* 2018), achieved notable accuracies focusing on three or less classes of either vowels or short words. In contrast, our study's ambition was to decode more complex linguistic content by examining the imagination of complete words, and using four classes, thereby presenting a distinct challenge. Abdulghani's work (Abdulghani *et al* 2023) also focused on the imagination of four complete words and got accuracies above 90%, but, as mentioned previously, they worked with runs containing 60 s of speech imagination, instead of the 2.5 s contained in the dataset we worked with. This strongly suggests that longer recordings of speech imagination could prove to be an effective strategy for enhancing the discriminability of the IS task.

Regarding the limitations of this study, some constraints need to be adressed. Firstly, the dataset utilized was notably restricted, characterized by a small participant pool (10 individuals) and a limited number of recordings per participant (approximately 60 runs for each imagined word). This constraint potentially limited the diversity and robustness of the dataset, which could have implications for the generalizability of our findings. Additionally, the relatively small sample size may have impacted the statistical power of our analyses, warranting cautious interpretation of the results.

Despite the growing interest demonstrated by the scientific community in constructing BCIs that utilize speech imagination as a strategy to generate recognizable signals through EEG, reflected in the significant increase in studies and databases published in recent years, we cannot discard the ingrained difficulties associated with this paradigm. These primarily arise from inherent limitations of EEG. Firstly, this technique is known for its challenging signal-to-noise ratio, making studies employing it arduous. Secondly, the limited quantity of samples commonly found in EEG databases (as was the case here), often due to the taxing nature of the acquisition process for participants, hinders the application of more advanced artificial intelligence methods such as deep neural networks. Nevertheless, in this study we were able to demonstrate the effectiveness of connectivity-based features in classifying IS brain signals. This has implications for enhanced IS recognition in BCI applications, as well as for improved neurophysiological signal analysis, and advancements in assistive technology and cognitive research. As more databases are published, including higher-quality datasets, it is foreseeable that in the coming years, we may obtain a more conclusive answer regarding the feasibility of determining the word a person is thinking based on the collected scalp information.

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Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://openneuro.org/datasets/ds003626/versions/2.1.2.

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